

IPSEC REGULATORY AUDIT & CRYPTOGRAPHIC HEALTH

# IPsec VPN Security Assessment & Framework Audit

Comprehensive evaluation against NIST SP 800-77 Rev. 1, RFC 8221, and NSA CNSA 2.0 cryptographic mandates

77  
GRADE B+

CONDITIONAL COMPLIANCE · ATTENTION REQUIRED

## Section 1: Scope & Regulatory Framework

Definition of audit benchmarks (NIST SP 800-77 Rev. 1, RFC 8221, NSA CNSA 2.0)

| AUDIT SCOPE & TARGET GATEWAYS |                      | ASSESSMENT TIMEFRAME                |                          |
|-------------------------------|----------------------|-------------------------------------|--------------------------|
| --                            |                      | --                                  |                          |
| REGULATORY BENCHMARK          | TITLE & JURISDICTION | MANDATED CRYPTOGRAPHIC REQUIREMENTS | AUDIT COMPLIANCE VERDICT |

## Section 2: Cryptographic Health Audit

Granular evaluation of symmetric ciphers (AES-GCM vs. CBC/3DES) and hashing integrity (SHA-256 vs. SHA-1/MD5)

**SYMMETRIC ENCRYPTION CIPHER**

**AES-256-GCM (AEAD)**

Operating Mode: Galois/Counter Mode (AEAD)  
Integrity Tag: 128-bit ICV (Authentication Tag)  
Key Length: 256-bit Symmetric Key

APPROVED

AEAD provides built-in authentication, completely neutralizing padding oracle vulnerabilities.

**HASHING & MESSAGE INTEGRITY**

**SHA-256 / HMAC-SHA2-256**

Digest Size: 256-bit Digest  
Collision Resistance: Verified (FIPS 180-4 Compliant)  
Deprecated Primitives: MD5 / SHA-1 Not Detected

APPROVED

Collision-resistant hashing adherence verified with zero deprecated digest algorithms detected.

KEY MANAGEMENT, PROTOCOL & STATE INTEGRITY

# Key Exchange & Protocol Integrity

Diffie-Hellman modulus size verification, post-quantum posture, sequence rollover safeguards, and anti-replay integrity

## Section 3: Key Management & Exchange Audit

Verification of Diffie-Hellman modulus size, post-quantum posture, and PFS rekey enforcement

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| <div><div>DIFFIE-HELLMAN MODULUS STRENGTH</div><div>APPROVED</div></div> <div><div>ECP-256 (DH Group 19)</div><div>256-bit Elliptic Curve (~3072-bit RSA Equivalent)</div><div>Meets NIST SP 800-57 discrete logarithm strength recommendations for enterprise confidentiality.</div></div> | <div><div>PERFECT FORWARD SECRECY (PFS) REKEY</div><div>CRITICAL NON-COMPLIANCE</div></div> <div><div>Disabled on Child SA</div><div>IKE SA: Active (ECP-256)   Penalty: -11 pts</div><div>Child SA rekeying without ephemeral DH exchange allows retroactive bulk decryption if long-term credentials leak.</div></div> | <div><div>POST-QUANTUM CRYPTOGRAPHIC POSTURE</div><div>TRANSITION REQUIRED</div></div> <div><div>Classical ECC (Pre-Quantum)</div><div>Target: Hybrid ML-KEM-768 (Kyber)</div><div>Advisable to adopt hybrid post-quantum key encapsulation per NSA CNSA 2.0 roadmap (2026-2030).</div></div> | <div><div>SA ROTATION LIFETIMES &amp; LIMITS</div><div>ALIGNED</div></div> <div><div>3600 seconds (1 Hour)</div><div>1,000 MB (1 GB Transfer Threshold)</div><div>Strict rekey intervals constrain cryptanalytic exposure window under high packet volumes.</div></div> |
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## Section 4: Protocol & State Integrity

Anti-replay window sizing, sequence rollover safeguards, and IKE mode security (e.g., IKEv1 Aggressive Mode detection)

|                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>ANTI-REPLAY WINDOW SIZING</div><div>ACTIVE &amp; VERIFIED</div></div> <div><div>64-Bit Sliding Window (RFC 4303)</div><div>Kernel state maintains strict sequence bitmap. Monitored session accepted 0 duplicate or injected frames.</div></div> | <div><div>SEQUENCE ROLLOVER SAFEGUARDS</div><div>HARDENED</div></div> <div><div>64-Bit Extended Sequence Numbers (ESN)</div><div>Protects against 2^32 sequence cycling attacks under 10Gbps+ multi-core line-rate traffic. Safe for &gt;500 years.</div></div> | <div><div>IKE MODE SECURITY &amp; DOWNGRADE</div><div>PROTECTED</div></div> <div><div>IKEv2 (Main Mode Equivalent)</div><div>IKEv1 Aggressive Mode NOT detected. Unencrypted identity hashing is prevented, neutralizing offline dictionary cracking.</div></div> |
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SIDE-CHANNEL AUDIT & COMPREHENSIVE VULNERABILITY REGISTER

# Side-Channel Audit & Threat Matrix

Traffic Flow Confidentiality (TFC) leakage, itemized CVSS vulnerability register, and post-remediation posture

Section 5: Side-Channel & Metadata Vulnerability Audit

Leakage score based on packet shape predictability, burstiness, and missing Traffic Flow Confidentiality (TFC) padding

SIDE-CHANNEL LEAKAGE SCORE

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MODERATE RISK

PACKET SHAPE PREDICTABILITY

HIGH PREDICTABILITY

Length clustering around 1420B and 180B leaks media application type.

BURSTINESS & IAT TIMING

MODERATE LEAKAGE

Regular 20ms inter-arrival intervals reveal active media codec timing.

TFC PADDING AUDIT

MISSING / DISABLED (-3 pts)

Activate random TFC padding (RFC 4303 §2.7) to obscure packet lengths.

Section 6: Comprehensive Threat & Vulnerability Matrix

Detailed register of discovered vulnerabilities, CVSS base/environmental scores, and audit remediation mandates

| VULN ID | VULNERABILITY & THREAT VECTOR | CVSS v3.1 BASE / ENV | SEVERITY | REMEDIATION MANDATE |
|---------|-------------------------------|----------------------|----------|---------------------|
|---------|-------------------------------|----------------------|----------|---------------------|

PROJECTED POST-REMEDIATION POSTURE

CURRENT SCORE77 / 100

PROJECTED GAIN+19 pts

PROJECTED SCORE96 / 100

PROJECTED GRADEGRADE A+

Executing the top two priority directives (enforcing PFS Diffie-Hellman Group 19 on Child SAs and migrating to X.509 certificates) eliminates 16 penalty points, promoting the deployment into a fully hardened state compliant with NIST SP 800-77 Rev 1.